

TEST PLAN

Auto-Uncoding Audit Trail When Coded Datum Is Edited

Impacted Area: Medical Coding / Uncoding Audit Trail

1. Document Control

Field	Details
Document Title	
Version	1.0
Author	Jeevan Paapagaari, Senior QA Engineer
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JIRA Ticket	
Related Fix	
Fix Version	
Feature Branch / Build	
Master	Dev Master / TBD
Stage5	Stage5 / TBD

2. Introduction & Objectives

2.1 Purpose

This Test Plan defines the testing strategy, scope, test cases, and execution approach for JIRA ticket EDC-42690. The ticket addresses a defect and feature gap in the Medical Coding module of EDC, wherein editing a coded datum on a form does not remove the associated *DatumCoding* record or generate an 'Auto Uncoding' audit trail entry. This plan ensures the fix is verified end-to-end, covering functional correctness, audit trail compliance, data integrity, and regression safety.

2.2 Background / Problem Statement

When a datum is updated on a form, the system creates a new datum record and marks the old record as disabled. If the old datum was medically coded:

- The new datum appears as Not Coded.
- The old (disabled) datum still shows as Coded.
- No 'Uncoding' audit trail entry is created, breaking regulatory traceability requirements.

The fix ensures:

- The old *DatumCoding* record is removed
- An Auto Uncoding event is written to *AuditValueChange_tbl* before removal
- The new audit message is distinct from manual uncoding
- *ChangedObjectDescription* retains the pre-edit verbatim value for compliance.

2.3 Scope

In Scope

- Medical coding workflow — code, uncode (manual and auto)
- Editing a coded form verbatim value that triggers system auto-uncode
- Audit entries in *AuditValueChange_tbl*: message format, 'By [user]' attribution, *ChangedObjectDescription*

- Coding record removal from DatumCoding_tbl for old/disabled datum
- Multi-user traceability (coded by one user, edited by another)
- Edge cases: no-op save, unrelated field edit, sequential edits, clearing value
- Permissions and role-based access checks
- Regression: Coding UI filters, eCRF audit log, Medical Coding audit log/report
- DB integrity: DatumCoding_tbl, AuditValueChange_tbl

Out of Scope

- Coding dictionary content correctness and terminology validation
- Migration test cases (ticket confirms: Migration Test Cases = No)
- API changes (ticket confirms: Will this require API changes = No)
- Performance load testing beyond sanity checks
- Infrastructure and deployment validation

2.4 Testing Objectives & Success Criteria

Objective	Success Criterion
Validate auto-uncoding on verbatim edit	• DatumCoding record removed + correct audit entry created on every verbatim edit
Validate manual coding/uncoding baseline	• All existing audit formats unchanged and correct
Validate audit message distinctness	• Auto uncode message differs from manual uncode message • No ambiguity
Validate multi-user attribution	• 'By [user]' reflects correct actor for each event
Validate no over-triggering	• No auto-uncode when unrelated field edited or value unchanged
Validate DB integrity	• No orphan coding records • Audit history persists even after hard-delete
Validate compliance reporting	• Audit UI/report shows correct sequence: Entered → Coded → Edited → Auto Uncoded

2.5 References

Reference	Detail
JIRA Ticket	EDC-42690: When a coded datum is updated/edited on a form, then maintain un-coding audit trail
Prerequisite Fix	EDC-42360: Uncoding audit logging bug fix (must be verified in this build)
Dev Plan	StudyDataUpdater.cs → Datum.cs: RemoveCoding() + RegisterAuditEvent() called on datum save
DB Tables	<i>DatumCoding_tbl, AuditValueChange_tbl</i>
Acceptance Criteria	Embedded in JIRA ticket 'Test Plan' field (manual code, manual uncode, auto-uncode message formats)

3. Test Strategy

3.1 Testing Levels & Applicability

Testing Level	Applicable?	Rationale
Unit Testing	Partial	Dev-owned: RemoveCoding() and RegisterAuditEvent() unit tests in Datum.cs / StudyDataUpdater.cs
Integration Testing	Yes	Validate interaction: form save → datum versioning → DatumCoding removal → audit event writes
System / Functional Testing	Yes	End-to-end UI + DB scenarios — core QA scope for this ticket
Regression Testing	Yes	Coding UI, eCRF audit log, Medical Coding audit log must not break with this change
UAT	No	Technical task; no UAT cycle required per ticket classification
Performance Testing	Sanity only	Verify save latency not degraded; no full load test needed

3.2 Testing Types

Testing Type	Priority	Coverage
Functional (UI)	Critical	Core scenarios 1–3 (coding, uncoding, auto-uncoding) plus edge cases 4–9
Database Validation	Critical	<i>DatumCoding_tbl</i> and <i>AuditValueChange_tbl</i> integrity after every scenario
Audit Trail / Compliance	Critical	Exact message formats, actor attribution, ChangedObjectDescription value
Regression	High	Coding UI state, eCRF audit log, Medical Coding report
Negative / Permission	High	Role-based access; blocked actions must not create audit side-effects
Boundary / Edge Case	Medium	Clear value (null/blank), sequential edits, no-op save, unrelated field edit
Performance Sanity	Low	Subjective check: saving response time not noticeably degraded

3.3 Testing Approach

- **Primary** approach: Manual functional testing (UI + DB verification) for this cycle.
- **Automation [Optional]:** API/UI automation recommended for Scenarios 1, 2, 3 (create datum → code → edit verbatim → verify audit). DB assertion automation: verify *DatumCoding_tbl* has no record for old datum + *AuditValueChange_tbl* has correct entry.
- **Risk-based prioritization:** Critical scenarios (TC-001 through TC-003) must pass before any regression suite is run.
- Two-user testing setup required (User A for eCRF data entry, User B for Medical Coding) to validate multi-user attribution.

3.4 Entry & Exit Criteria

Level	Entry Criteria	Exit Criteria
System / Functional	• Build 1.0.0.886 deployed on QA environment • Test data (study, form, coding dictionary) configured • Two test user accounts available	• All Critical & High test cases executed. • 0 open Critical/High defects. • Audit messages match exact formats
Regression	• Core scenarios (TC-001–TC-003) all passed	• No regression in coding UI, eCRF audit log, or Medical Coding report. • All Medium test cases passed

Level	Entry Criteria	Exit Criteria
DB Validation	Scenarios 2 and 3 executed successfully in UI	- DatumCoding_tbl contains no orphan records. - AuditValueChange_tbl contains all required entries with correct formats
Sign-off	- All test cases were executed. - Defect log reviewed	- QA sign-off obtained. - go/no-go checklist approved by Product Owner and QA Lead

4. Test Environment & Infrastructure

4.1 Environment Matrix

Environment	Purpose	Build	DB Access	Owner
Dev	Developer unit/integration testing	1.0.0.886 (dev-bran-09)	Yes	Dev Team
Feature Branch	Primary QA functional + regression testing	1.0.0.886	Yes	QA Team
Dev Master	Primary QA functional + regression testing		Yes	QA Team
Staging	Pre-prod smoke + regression	Promoted from QA	Limited	DevOps
Production	Not in scope for this plan	N/A	No	Ops

4.2 Tools & Frameworks

Category	Tool / Framework
Test Management	JIRA (tickets, test details) / Confluence (Test Plan documentation)
Test Execution	Manual execution via EDC UI + SQL queries against QA DB
DB Access	SQL Server Management Studio / DBeaver (access to DatumCoding_tbl, AuditValueChange_tbl)
Bug Tracking	JIRA — defects logged against EDC-42690 or as linked issues

4.3 Test Data Requirements

- Study with at least one form containing a variable linked to a Coding Dictionary.
- At least one subject/record with data entry access.
- User A: eCRF data entry role (can enter and edit form data).
- User B: Medical Coding role (can code/uncode datums, cannot edit eCRF data directly).
- Access to QA DB: DatumCoding_tbl, AuditValueChange_tbl, and datum table(s) to confirm disabled/new datum records.
- Test data must NOT use real patient data. Use synthetic/anonymized study data in the QA environment.
- The QA engineer must create test data from scratch

4.4 Environment Setup Checklist

#	Setup Task	Owner	Done?
1	Deploy build 1.0.0.886 on QA environment (branch: dev-bran-09)	Mr. Bhavya	Yes
2	Confirm EDC-42360 fix is included in this build (prerequisite for audit trail)	Mr. Jeevan	Yes
3	Create / verify study with a form variable linked to a Coding Dictionary	Mr. Jeevan	NTD

#	Setup Task	Owner	Done?
4	Create User A (eCRF data entry) and User B (Medical Coding) test accounts	Mr. Jeevan	NTD
5	Verify DB access to DatumCoding_tbl and AuditValueChange_tbl on QA environment	Mr. Jeevan	Yes
6	Confirm eCRF audit log UI and Medical Coding audit log / report are accessible	Mr. Jeevan	NTD
7	Smoke test: basic form entry and manual coding functions work (pre-test sanity)	Mr. Jeevan	NTD

5. Test Scope & Coverage Matrix

5.1 Feature-to-Test-Type Coverage Matrix

Feature Area	Functional	DB Valid.	Audit Trail	Regression	Negative	Perf.
Manual Coding Audit Trail	✓	✓	✓	✓	-	-
Manual Uncoding Audit Trail	✓	✓	✓	✓	-	-
Auto-Uncoding on Verbatim Edit (CORE)	✓	✓	✓	✓	✓	✓
Multi-User Attribution	✓	✓	✓	-	-	-
Sequential Edits (no duplicate auto-uncode)	✓	✓	✓	-	-	-
No-op Save / Unrelated Field Edit	✓	✓	-	-	✓	-
Clear/Null Verbatim (edge case)	✓	✓	✓	-	-	-
Permissions / Role Access	-	-	-	-	✓	-
Coding UI / Report Regression	-	-	✓	✓	-	-

5.2 Risk-Based Test Prioritization

Test Case	Description	Risk Level	Priority
TC-001	Manual coding creates correct audit entry	High	High
TC-002	Manual uncoding removes coding + correct audit	High	High
TC-003	Auto-uncoding on verbatim edit — CORE SCENARIO	Critical	Critical
TC-004	Multi-user attribution (coded by B, edited by A)	High	High
TC-005	Sequential edits — no duplicate auto-uncode entries	Medium	High
TC-006	Edit unrelated field — no auto-uncoding triggered	Medium	Medium
TC-007	No-op save — no auto-uncoding triggered	Medium	Medium
TC-008	Clear verbatim value — auto-uncoding fires correctly	Medium	Medium
TC-009	Permissions / role check — no side effects from blocked actions	Low	Low
TC-010	DB integrity & regression checks	Critical	Critical

5.3 Requirements Traceability Matrix

Requirement	Source	Covered By
Auto-uncoding audit entry written when verbatim edited	EDC-42690 AC	TC-003, TC-004, TC-005, TC-008, TC-010

Requirement	Source	Covered By
DatumCoding record removed for old datum on verbatim edit	EDC-42690 AC	TC-003, TC-005, TC-008, TC-010
Auto uncode message distinct from manual uncode message	EDC-42690 AC	TC-003, TC-004
ChangedObjectDescription holds pre-edit verbatim value	EDC-42690 AC	TC-003, TC-008, TC-010
Manual coding audit format: Term [...] : Coded By [user] at [dt]	EDC-42690 AC	TC-001, TC-004, TC-010
Manual uncoding audit format: Term [...] : Uncoded By [user] at [dt]	EDC-42690 AC	TC-002, TC-010
No auto-uncode when non-verbatim field edited or no-op save	EDC-42690 AC	TC-006, TC-007
Unauthorized users cannot trigger auto-uncoding	Security / compliance	TC-009
EDC-42360 fix: audit trail present even when coding row hard-deleted	EDC-42360 (prerequisite)	TC-002, TC-003, TC-010

6. Test Case Structure

Each test case below maps to a scenario from the JIRA RoVo test plan. All test cases include: ID, Title, Pre-conditions, Steps, Expected Results, Actual Result (blank), Status (blank), Priority, Test Type.

TC-001	Manually code a datum (baseline)	High	Positive
Field	Details		
Preconditions	1. A form exists with a variable linked to a Coding Dictionary. 2. User A and User B accounts are available. 3. QA DB access is available.		
Test Steps	1. Create or identify a form containing a variable linked to a Coding Dictionary. 2. As User A, enter a verbatim value (e.g., 'Medication1') into the field and save the form. 3. Navigate to the Coding section in EDC. 4. Select the variable/datum for which data was entered. 5. Code the datum by selecting an appropriate coding term. 6. Query AuditValueChange_tbl for the coding event. 7. Query DatumCoding_tbl for the coding record.		
Expected Result	1. The datum is marked as 'Coded' in the Coding UI. 2. AuditValueChange_tbl has an entry with TargetObjectDescription = 'Term [name_of_term] : Coded By [user] at [datetime]'. 3. DatumCoding_tbl contains a record for this coded datum.		
Actual Result			
Status	[] Pass [] Fail [] Blocked [] N/A		
Traces To	REQ: Manual coding audit format (EDC-42690 AC)		

TC-002	Manually uncode a datum (baseline)	High	Positive
Field	Details		

Preconditions	TC-001 completed: datum is coded. QA DB access available.
Test Steps	1. Navigate to Coding in the EDC application. 2. Select the datum that is already coded (from TC-001). 3. Click 'Uncode' to manually uncode the datum. 4. Query AuditValueChange_tbl for the uncoding event. 5. Query DatumCoding_tbl to verify the coding record is removed.
Expected Result	1. The coding is removed from the datum in the Coding UI. 2. AuditValueChange_tbl has an entry with TargetObjectDescription = 'Term [name_of_term] : Uncoded By [user] at [datetime]'. 3. DatumCoding_tbl no longer contains the coding record for this datum (hard-deleted, per existing behavior).
Actual Result	
Status	[] Pass [] Fail [] Blocked [] N/A
Traces To	REQ: Manual uncoding audit format; REQ: EDC-42360 fix (audit persists after hard-delete)

TC-003	AUTO-UNCODING on verbatim edit — CORE SCENARIO	Critical	Positive
Field	Details		
Preconditions	TC-001 completed: datum is coded. Do NOT manually uncode. QA DB access available.		
Test Steps	1. As User A, open the same form where the coded datum exists. 2. Edit the verbatim value of the coded datum (e.g., change 'Medication1' to 'Medication2'). 3. Save the form. 4. Verify the save succeeds (no errors). 5. Check the Coding UI: confirm new datum shows 'Not Coded'. 6. Confirm old/disabled datum does not appear as 'Coded' in any user-facing view. 7. Query DatumCoding_tbl: confirm no coding record exists for the old/disabled datum. 8. Query AuditValueChange_tbl: confirm auto-uncoding audit entry.		
Expected Result	1. Save succeeds without errors. 2. New datum (post-edit) shows 'Not Coded' in Coding UI. 3. DatumCoding_tbl has NO coding record for the old/disabled datum. 4. AuditValueChange_tbl has entry with: - TargetObjectDescription = 'Term [term_name]: Auto Uncoded - Verbatim updated By [user] at [datetime]' - ChangedObjectDescription = 'Medication1' (the prior verbatim value) 5. Auto-uncode message is DISTINCT from manual uncode message (no 'Uncoded By' — must say 'Auto Uncoded - Verbatim updated').		
Actual Result			
Status	[] Pass [] Fail [] Blocked [] N/A		
Traces To	REQ: Auto-uncoding audit entry; REQ: DatumCoding removal; REQ: ChangedObjectDescription; REQ: Distinct message		

TC-004	Multi-user traceability — 'By [user]' attribution correctness	High	Positive
Field	Details		
Preconditions	User A (data entry) and User B (coder) accounts are active and accessible.		
Test Steps	1. As User A, enter verbatim value (e.g., 'Medication1') on the form and save. 2. As User B, navigate to Coding and code that datum (select a term). 3. As User A, open the same form and edit the coded verbatim (change 'Medication1' to 'Medication2'). 4. Save the form. 5. Query AuditValueChange_tbl for both the coding event and the auto-uncoding event.		

Expected Result	1. Coding audit entry: 'Term [name_of_term] : Coded By [User B] at [datetime]' 2. Auto-uncoding audit entry: 'Term [term_name]: Auto Uncoded - Verbatim updated By [User A] at [datetime]' 3. 'By [user]' correctly reflects the actor who performed the respective action.
Actual Result	
Status	[] Pass [] Fail [] Blocked [] N/A
Traces To	REQ: 'By [user]' attribution for coding; REQ: 'By [user]' attribution for auto-uncoding

TC-005	Sequential edits — no duplicate auto-uncode entries	High	Negative
Field	Details		
Preconditions	Coded datum exists (TC-001 completed). Do NOT manually uncode.		
Test Steps	1. Edit the coded verbatim and save (triggers auto-uncoding, removes coding). 2. Without re-coding, edit the same verbatim again and save. 3. Query AuditValueChange_tbl for auto-uncoding entries. 4. Query DatumCoding_tbl for any orphan records.		
Expected Result	1. First edit: one auto-uncoding audit entry created; DatumCoding record removed. 2. Second edit (no coding exists): no additional auto-uncoding entries generated. 3. No duplicate or orphan records in DatumCoding_tbl. 4. No application errors.		
Actual Result			
Status	[] Pass [] Fail [] Blocked [] N/A		
Traces To	REQ: No duplicate auto-uncode when no coding exists; REQ: No orphan DatumCoding records		

TC-006	Edit unrelated field — no auto-uncoding triggered	Medium	Negative
Field	Details		
Preconditions	1. Form has at least two fields. One datum is coded. 2. Second field is not linked to a Coding Dictionary.		
Test Steps	1. Open the form containing the coded datum. 2. Change the value of a DIFFERENT field (not the coded verbatim variable). 3. Save the form. 4. Query DatumCoding_tbl and AuditValueChange_tbl.		
Expected Result	1. The coded datum remains coded; its DatumCoding record is intact. 2. No 'Auto Uncoded - Verbatim updated' audit entry created for the coded datum. 3. No errors on save.		
Actual Result			
Status	[] Pass [] Fail [] Blocked [] N/A		
Traces To	REQ: No auto-uncode when non-verbatim field edited		

TC-007	No-op save — open form and save without changing verbatim	Medium	Boundary
Field	Details		
Preconditions	Coded datum exists on a form.		
Test Steps	1. Open the form containing the coded datum. 2. Do not change the coded verbatim value.		

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	3. Click Save. 4. Query DatumCoding_tbl and AuditValueChange_tbl.
Expected Result	1. Coding remains intact; DatumCoding record unchanged. 2. No auto-uncoding event triggered. 3. No new auto-uncoding audit entry in AuditValueChange_tbl. 4. No new datum record created (or if system creates one, coding is preserved since value did not change).
Actual Result	
Status	[] Pass [] Fail [] Blocked [] N/A
Traces To	REQ: No auto-uncode on no-op save

TC-008	Clear verbatim value (null/blank) — auto-uncoding fires	Medium	Edge
Field	Details		
Preconditions	Coded datum exists. System allows the verbatim value to be cleared/set to blank. [NEEDS CLARIFICATION: confirm if clearing verbatim is permitted by validation rules]		
Test Steps	1. Open the form and clear the coded verbatim value (set to blank/null). 2. Save the form. 3. Query DatumCoding_tbl and AuditValueChange_tbl.		
Expected Result	1. Auto-uncoding is triggered (verbatim value changed). 2. DatumCoding record for the old datum is removed. 3. AuditValueChange_tbl entry created with: - TargetObjectDescription = 'Term [term_name]: Auto Uncoded - Verbatim updated By [user] at [datetime]' - ChangedObjectDescription = prior non-empty value (e.g., 'Medication1').		
Actual Result			
Status	[] Pass [] Fail [] Blocked [] N/A		
Traces To	REQ: Auto-uncoding on verbatim change; REQ: ChangedObjectDescription contains prior value		

TC-009	Permissions / role check — blocked actions produce no side effects	Low	Negative
Field	Details		
Preconditions	A user account WITHOUT eCRF edit permissions and a user WITHOUT Coding permissions are available.		
Test Steps	1. Log in as a user without eCRF edit permissions. 2. Attempt to edit the coded verbatim value and save. 3. Log in as a user without Coding permissions. 4. Attempt to code or uncode a datum.		
Expected Result	1. User without eCRF edit permission cannot save changes — action is blocked by the system. 2. No auto-uncoding event created; DatumCoding_tbl and AuditValueChange_tbl unchanged. 3. User without Coding permission cannot code/uncode — action is blocked. 4. No coding audit side effects.		
Actual Result			
Status	[] Pass [] Fail [] Blocked [] N/A		
Traces To	REQ: Unauthorized users cannot trigger auto-uncoding		

TC-010	Data integrity & regression — DB + UI validation suite	Critical	Positive
Field	Details		

Preconditions	TC-001 through TC-003 (minimum TC-002 and TC-003) executed. QA DB access available. Audit log UI and Medical Coding report accessible.
DB Checks	1. No orphan coding records: DatumCoding_tbl contains no entries for disabled datums after auto-uncoding. 2. Audit history presence: AuditValueChange_tbl has entries for: - Manual coding events (TC-001) - Manual uncoding events (TC-002) - Auto-uncoding events (TC-003, TC-005, TC-008) 3. Correct message formats verified in AuditValueChange_tbl. 4. ChangedObjectDescription stores prior verbatim value for auto-uncode rows. 5. Correct linkage: audit entries reference correct datum/subject/form context.
UI / Report Checks	1. Coding UI: coded/not-coded states update correctly after edit/save. 2. No stale 'Coded' state remains on disabled datum in user-facing views. 3. eCRF audit log shows 'data entered' and 'edited' events in correct sequence. 4. Medical Coding audit log shows: Coded → [Edited in eCRF] → Auto Uncoded - Verbatim updated (against original verbatim). 5. New verbatim has no coding history in Medical Coding log (it was never coded).
Expected Result	All DB and UI checks pass. No orphan records. Audit trail is complete and immutable.
Actual Result	
Status	[] Pass [] Fail [] Blocked [] N/A
Traces To	All requirements; compliance traceability requirement

7. Defect Management (Severity & Priority Classification Matrix)

Severity	Definition	Example (this ticket)	Resolution SLA
Critical (S1)	System crash, data corruption, incorrect audit entry that breaks compliance	• Auto-uncoding audit entry not written. • DatumCoding record not removed. • wrong user in 'By [user]'	Same day
High (S2)	Core functionality broken but workaround exists; incorrect message format	• Auto-uncode triggered on no-op save. • Duplicate audit entries on sequential edits	24 hours
Medium (S3)	Partial functionality affected; minor data inconsistency	• Coding UI shows stale 'Coded' state after auto-uncode. • Wrong ChangedObjectDescription value	48 hours
Low (S4)	Cosmetic or minor usability issue; no data impact	• Audit log timestamp format inconsistency; UI label wording	72 hours

8. Roles & Responsibilities (RACI Matrix)

Activity	QA Lead	QA Engineer	Dev Team	Product Owner	DevOps	Security
Test Plan creation & review	A/R	C	C	I	-	-
Test environment setup & deployment	A	C	C	I	R	-
Test data preparation	A	R	C	-	-	-
Test case execution (functional)	A	R	I	I	-	-
DB validation queries	A	R	C	-	-	-

Activity	QA Lead	QA Engineer	Dev Team	Product Owner	DevOps	Security
Defect triage & prioritization	R/A	C	R	C	-	-
Defect fix & unit testing	I	I	R/A	-	-	-
Regression execution	A	R	I	I	-	-
Permission/security checks	A	R	C	-	-	C
QA sign-off & go/no-go decision	R/A	C	C	A	-	-

R = Responsible | A = Accountable | C = Consulted | I = Informed

9. Test Execution Plan

9.1 Execution Timeline (For every environment: Feature Branch, Dev Master and Stage5)

Phase	Activities	Who	Sprint Day	Exit Gate
Setup	Deploy build; configure test study/users; verify DB access; smoke test	QA + DevOps	Day 1	Smoke pass
Critical Path	Execute TC-001, TC-002, TC-003 (manual coding, uncoding, auto-uncode). Run DB validation. Log defects if any.	QA Engineer	Day 1	0 Critical defects
Extended Scenarios	Execute TC-004 (multi-user), TC-005 (sequential edits), TC-006 (unrelated field), TC-007 (no-op)	QA Engineer	Day 1	0 High defects
Edge Cases	Execute TC-008 (clear value), TC-009 (permissions)	QA Engineer	Day 1	All Medium executed
Regression	TC-010 DB + UI regression. Coding UI, eCRF audit log, Medical Coding report checks.	QA Engineer	Day 1	No regression
Defect Retest	Retest any fixed defects from above phases	QA Engineer	Day 1	All defects closed
Sign-off	QA summary report; go/no-go checklist; stakeholder approval	QA Lead + PO	Day 1	Sign-off obtained

9.2 Daily Reporting Cadence

- End-of-day summary: test cases executed, passed, failed, blocked; new defects logged.
- Defect status update: open defect count by severity, ETA for fixes from Dev.
- Build change notification: QA Lead notified of any new build deployment before continuing.
- Showstopper escalation: immediate Slack/email to QA Lead + Dev Lead if TC-003 or TC-010 fails critically.

9.3 Showstopper / Blocker Escalation Path

Scenario	Escalation
TC-003 fails (auto-uncoding not working — core scenario)	Immediate: QA Lead → Dev Lead → Product Owner. Testing paused until fix deployed.
DB access unavailable (cannot validate DatumCoding_tbl / AuditValueChange_tbl)	QA Lead → DevOps. Testing blocked until DB access restored.
Build not deployed or smoke test fails	QA Lead → DevOps. Delay execution until stable build confirmed.
Prerequisite fix (EDC-42360) not present in this build	Immediate: QA Lead → Dev Lead. Build is invalid for this test plan.

10. Risks & Mitigation

Risk	Likelihood	Impact	Mitigation Plan	Owner
EDC-42360 prerequisite fix not in build 1.0.0.886	Medium	Critical	- Verify EDC-42360 during environment setup - Fail the build if missing.	Jeevan
DB access restricted in QA environment	Low	High	- Confirm DB access on Day 1. - Request DevOps to grant access proactively.	Jeevan
Auto-uncoding triggers on no-op save (over-triggering bug)	Medium	High	- TC-007 specifically tests this. - Immediate Critical defect if observed.	Jeevan
ChangedObjectDescription stores wrong / empty value	Medium	Critical	- Verified in TC-003 and TC-008. - DB query explicitly checks this field.	Jeevan
Regression: existing manual uncode behavior broken	Medium	High	- TC-002 and TC-010 cover manual uncode regression. - Run before sign-off.	Jeevan
Concurrency: two users editing same form simultaneously	Low	Medium	- Sanity checks only. - Log as known risk if concurrency is not supported.	Bhavya
Test environment instability causing false failures	Low	Medium	- Use stable QA environment. - Coordinate deployments with DevOps.	DevOps Team

11. Metrics & Reporting

11.1 Key Performance Indicators (KPIs)

KPI	Formula / Definition	Target for Sign-off
Test Case Execution Coverage	$(\text{Executed TCs} / \text{Total TCs}) \times 100$	100%
Test Pass Rate	$(\text{Passed TCs} / \text{Executed TCs}) \times 100$	$\geq 95\%$ (all Critical = 100%)
Critical Defect Count (open)	Open defects with Severity = Critical	0 at sign-off
High Defect Count (open)	Open defects with Severity = High	0 at sign-off
Defect Density	Total defects found / Total TCs executed	[NEEDS CLARIFICATION: project baseline]
Defect Reopen Rate	$(\text{Reopened defects} / \text{Total closed defects}) \times 100$	$< 10\%$
Escaped Defects (post-release)	Defects found in Staging or Production after QA sign-off	0 for Critical/High

11.2 Status Report Template

Field	Value
Date	[Date]
Build Under Test	1.0.0.886 (dev-bran-09)
TCs Planned Today	
TCs Executed	
Passed / Failed / Blocked	
New Defects Logged	

Field	Value
Open Defects (by severity)	Critical: _ High: _ Medium: _ Low: _
Blockers / Risks	

11.3 Definition of Done (DoD) — QA Sign-off

- All 10 test cases were executed.
- All Critical and High test cases passed.
- 0 open Critical defects.
- 0 open High defects.
- DB validation (TC-010) passed: no orphan coding records; all audit entries present with correct formats.
- Regression checks passed: Coding UI, eCRF audit log, Medical Coding report.
- All defects logged, triaged, and either closed or formally accepted as deferred.
- QA summary report delivered to stakeholders.
- Go/No-Go checklist signed off by QA Lead and Product Owner.

12. Sign-Off Criteria & Approval

12.1 Go / No-Go Checklist

#	Criterion	Status	Notes
1	Build 1.0.0.886 on QA environment with EDC-42360 fix included	☐ Yes ☐ No	
2	TC-001 (Manual Code) — PASSED	☐ Yes ☐ No	
3	TC-002 (Manual Uncode) — PASSED	☐ Yes ☐ No	
4	TC-003 (Auto-Uncode Core) — PASSED with all 5 acceptance criteria met	☐ Yes ☐ No	
5	TC-004 through TC-009 all executed and PASSED or formally accepted	☐ Yes ☐ No	
6	TC-010 DB integrity — PASSED: no orphan records; all audit entries present	☐ Yes ☐ No	
7	Zero (0) open Critical defects	☐ Yes ☐ No	
8	Zero (0) open High defects	☐ Yes ☐ No	
9	Regression checklist for Coding UI and audit logs — PASSED	☐ Yes ☐ No	
10	QA Summary Report delivered to Product Owner	☐ Yes ☐ No	

END OF TEST PLAN — EDC-42690 v1.0